Creating an agent

4 minutes

Agents are *applications* built with generative AI models. Agentic AI moves beyond one-off prompts and instead defines a consistent, workflow-like behavior that can be reused across apps, experiences, and services.

An agent in Microsoft Foundry is a packaged, reusable AI component that brings together three things:

- **A model**: the generative AI model the agent uses for reasoning (for example, GPT-4.1)
- **Instructions**: the system prompt that defines the agent's role, behavior, style, constraints, and output rules
- **Tools**: the actions the agent can take

Agents can:

- Call external tools (APIs, functions, retrieval) automatically
- Break goals into structured steps
- Maintain working memory during a conversation
- Process user input, decide actions, and generate structured outputs

Create an agent in Foundry portal

To create an agent in Foundry, you can start by exploring a model or just go straight to agent development. In Foundry portal, creating an agent looks similar at first to testing a model in the playground.

1. Choose the model your agent uses.
2. Write the system instructions, such as "You're a helpful scheduling assistant who returns answers in concise bullet points."

What sets the agent apart from using the model alone is the addition of tools, which allow the model to act on information and knowledge, which grounds the model with information.

Tools = *actions*.
Knowledge = *context*.

Add Tools

Tools in Foundry allow a model to perform actions by calling external systems. They represent **callable capabilities** such as searching the web, querying a database, or using an MCP server.

When enabled in the model playground, the model can inspect available tools, then call them when relevant to a user request. Examples of tools include:

- Code Interpreter (data analysis, file handling)
- Using knowledge sources
- Custom functions or APIs

Tools allow the model to:

- Take real actions (read/write files, search, update systems)
- Execute workflows
- Integrate into enterprise systems

In Foundry, tools form the basis for action-taking agents. They can be configured centrally using the **Foundry Tool Catalog**, where you can discover and manage tools.

Add Knowledge

Knowledge allows the model to **access and retrieve external content** (your documents, datasets, internal sites) through retrieval-augmented generation (RAG).

Knowledge in Foundry refers to **documents or datasets** provided to the model so it can retrieve highly relevant context during generation. Data can include internal PDFs, SharePoint content, Azure Storage files, and multi-source knowledge bases.

In the playground, Foundry uses retrieval pipelines to:

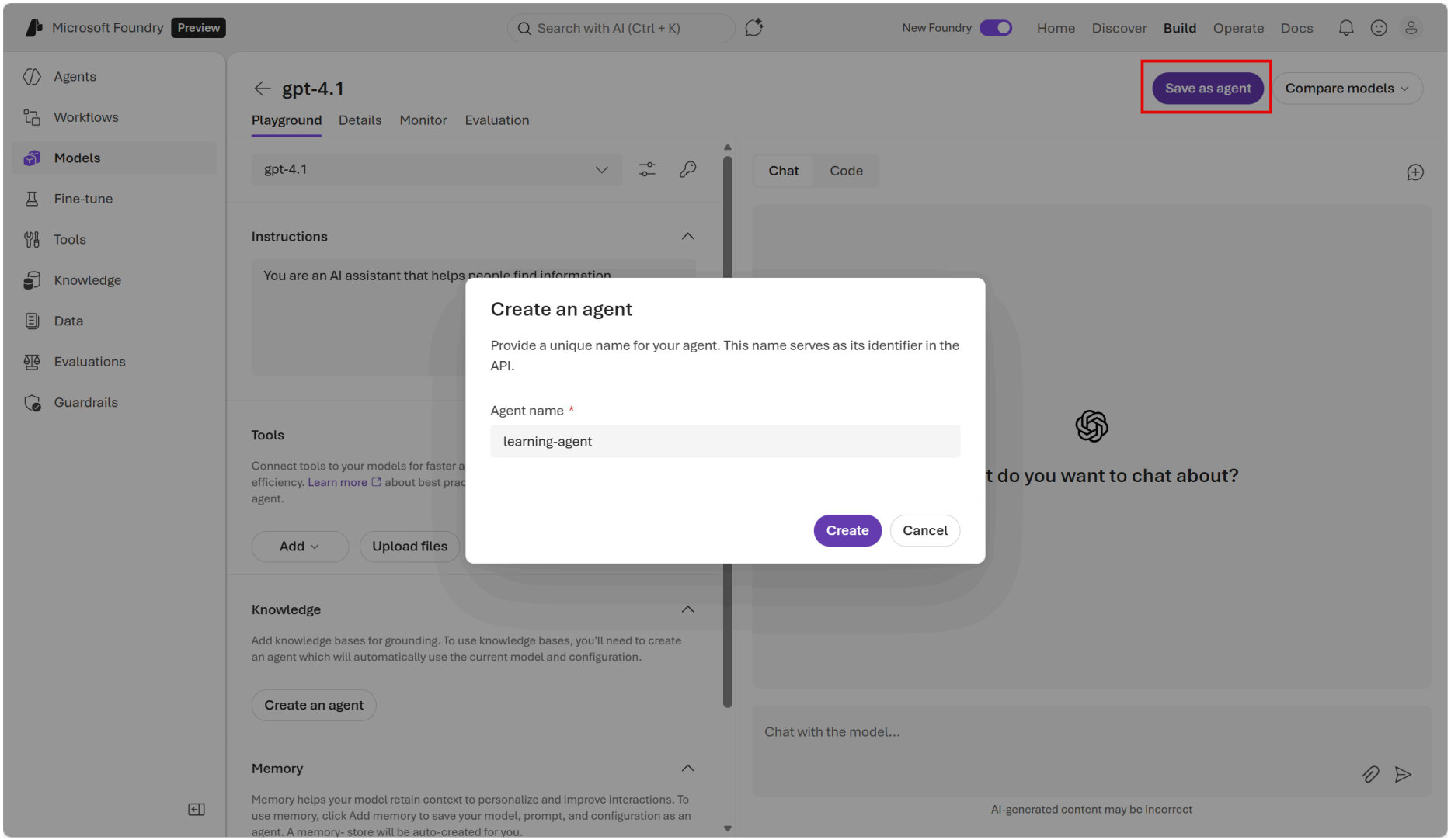
1. **Ingest + index** your content
2. **Search + ground** responses
3. Make answers more accurate, traceable, and domain-specific

Agents rely heavily on knowledge when answering domain-specific questions. When knowledge is used, the response includes a citation for the knowledge store the agent used.

Knowledge enables:

- Document-grounded Q&A
- Context-rich assistance
- Enterprise-safe retrieval

In the Foundry portal, you can save your model, instructions, and tools as an agent. You can continue to test and refine your agent in the Playground.


Microsoft Foundry
Preview
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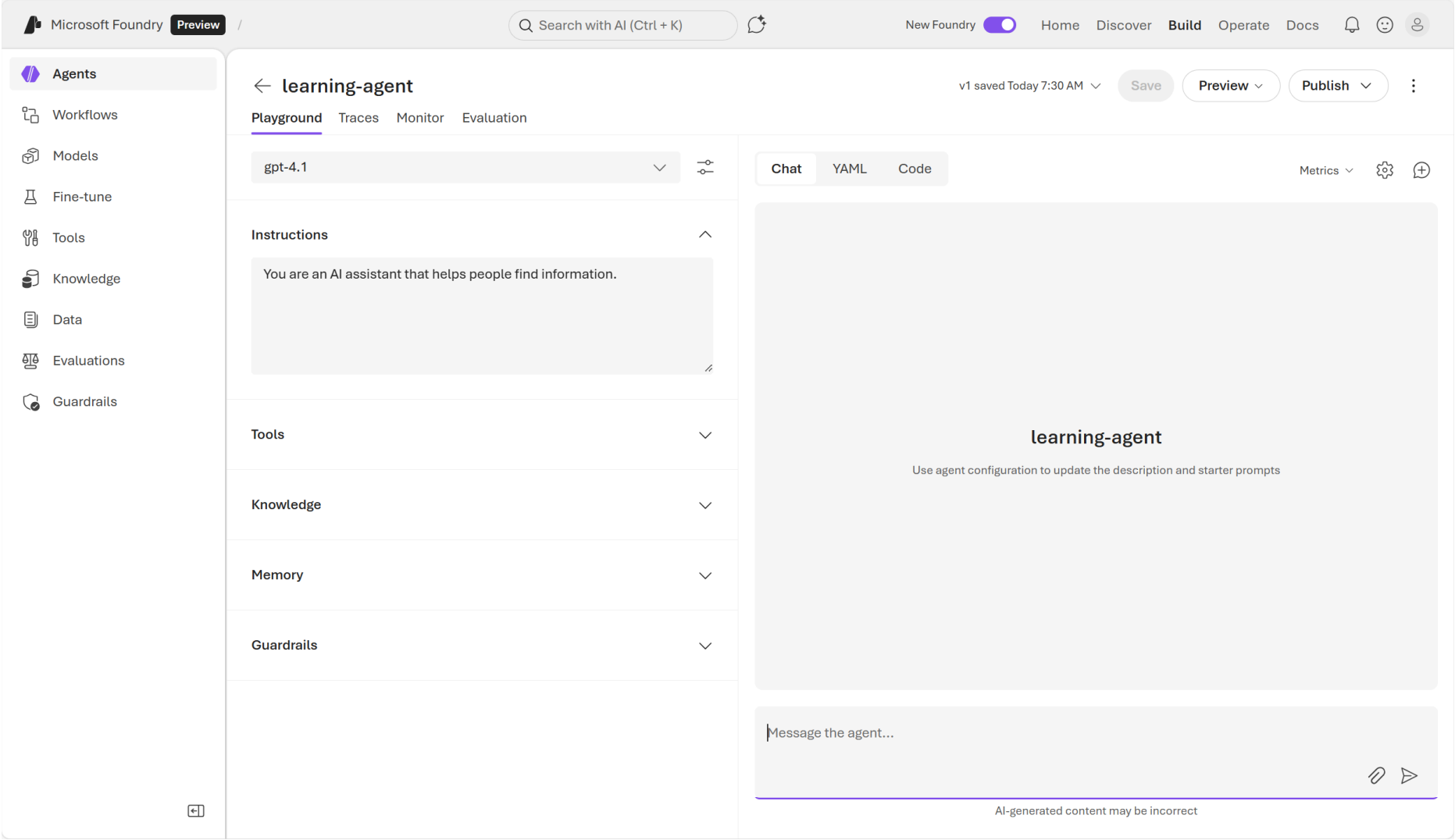
Docs
Agents
Workflows
Models
Fine-tune
Tools
Knowledge
Data
Evaluations
Guardrails
gpt-4.1
Save as agent
Compare models
Playground
Details
Monitor
Evaluation
gpt-4.1
Chat
Code
Instructions
You are an AI assistant that helps people find information
Create an agent
Provide a unique name for your agent. This name serves as its identifier in the API.
Agent name *
learning-agent
Create
Cancel
Tools
Connect tools to your models for faster a
efficiency. Learn more about best prac
agent.
Add
Upload files
t do you want to chat about?
Knowledge
Add knowledge bases for grounding. To use knowledge bases, you'll need to create an agent which will automatically use the current model and configuration.
Create an agent
Chat with the model...
Memory
Memory helps your model retain context to personalize and improve interactions. To use memory, click Add memory to save your model, prompt, and configuration as an agent. A memory- store will be auto-created for you.
AI-generated content may be incorrect



Using an agent

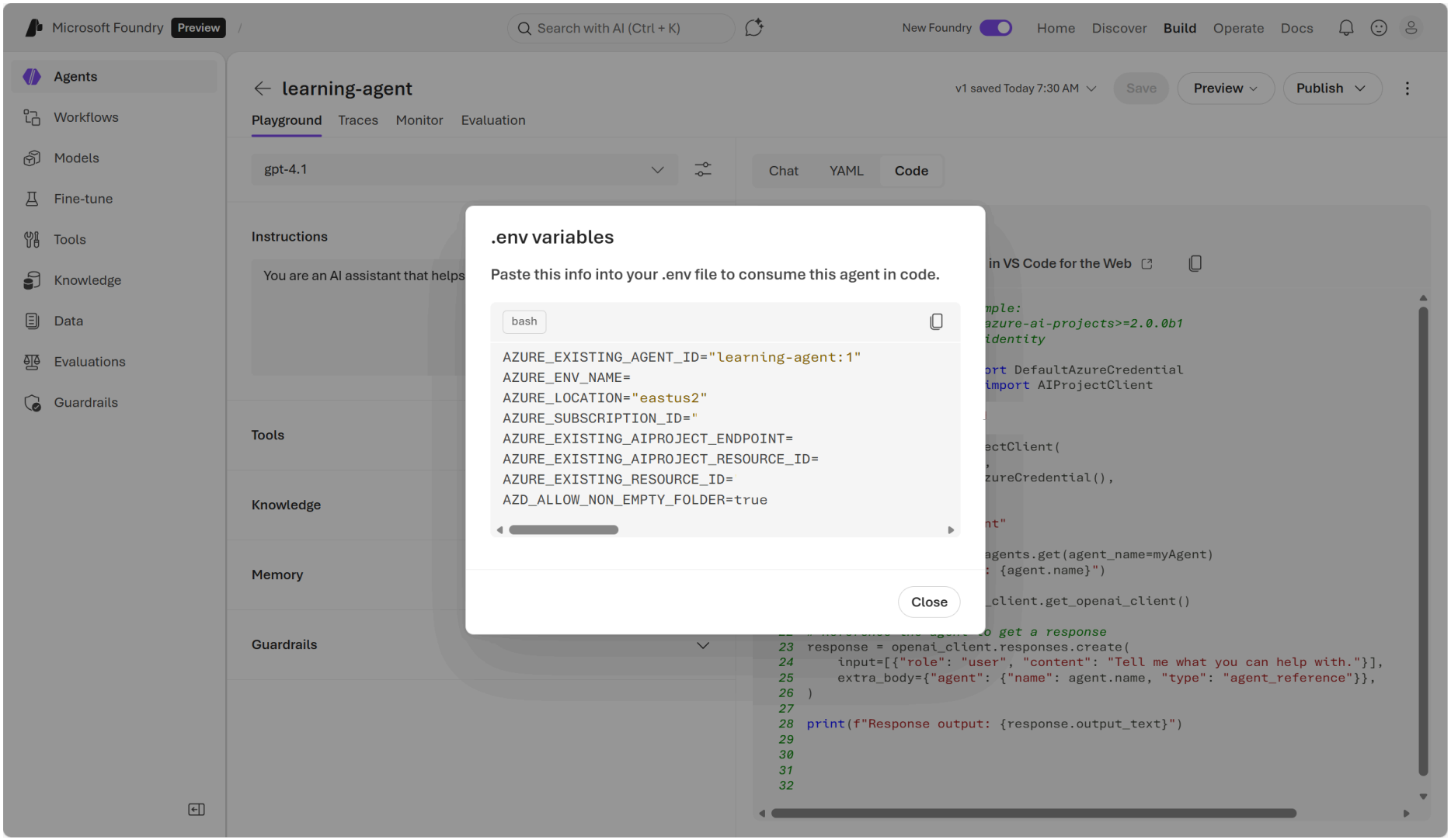
You can use an agent from a client application by using the **Foundry Projects SDK** to connect to the project and call it from a clie using the **Project API**.

The Project API enables you to:

- Integrate agents into web apps, bots, or backend workflows
- Orchestrate multi-step tasks
- Pass structured inputs or tool calls
- Run agents at scale with your Foundry deployments

Create a client application for an agent

To call the agent programmatically using Foundry's Project API, you need the `agent-id` of your agent. You can find the `agent-id` in the Playground view of the agent when you select the *code* view and open the *.env variables*.



Let's take a look at a Python code sample to use an agent:

Python

```python
# Before running the sample, install the packages:
#    pip install --pre azure-ai-projects>=2.0.0b1
#    pip install azure-identity
```

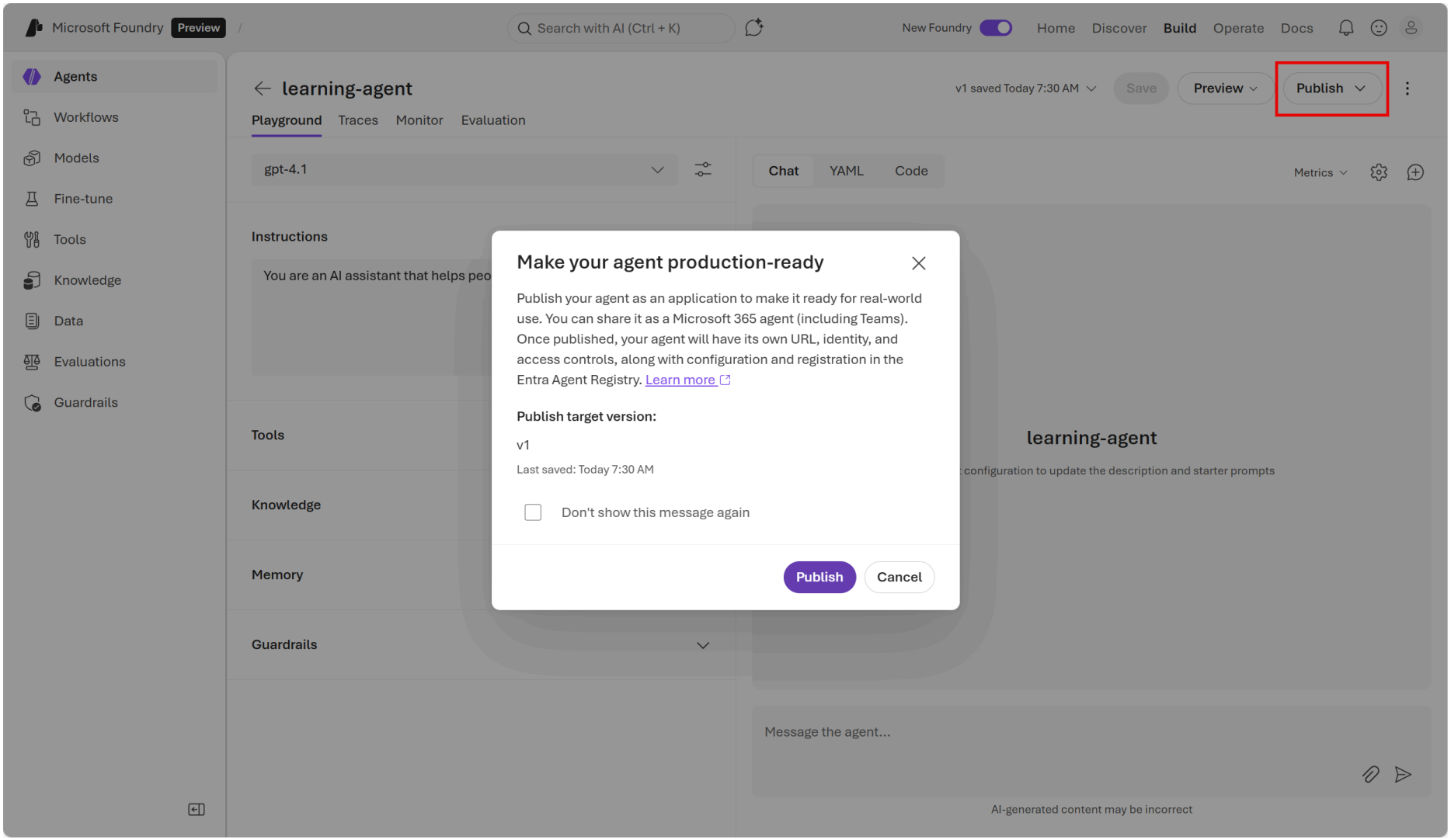
```python
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient

myEndpoint = "https://<resource>.services.ai.azure.com/api/projects/<resource-name>"

project_client = AIProjectClient(
    endpoint=myEndpoint,
    credential=DefaultAzureCredential(),
)

myAgent = "learning-agent"
# Get an existing agent
agent = project_client.agents.get(agent_name=myAgent)
print(f"Retrieved agent: {agent.name}")

openai_client = project_client.get_openai_client()

# Reference the agent to get a response
response = openai_client.responses.create(
    input=[{"role": "user", "content": "Tell me what you can help with."}],
    extra_body={"agent": {"name": agent.name, "type": "agent_reference"}},
)

print(f"Response output: {response.output_text}")
```

Publishing an agent

Publishing an agent changes it from a saved development asset to a managed Azure resource with a dedicated endpoint. A published agent can be shared without exposing the Foundry project or source code. You can publish an agent through the Foundry portal.

> **Note**

You don't need to publish an agent in order to use its endpoint. However, when you use the unpublished endpoint, you're essentially using the agent as a development asset inside your Foundry project. That is great for iteration, but not ideal for distribution, governance, or stable integrations. After you publish an agent, your agent gets its own unique endpoint that is different from the Foundry project endpoint.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Search with AI (Ctrl + K)
New Foundry
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Docs
Agents
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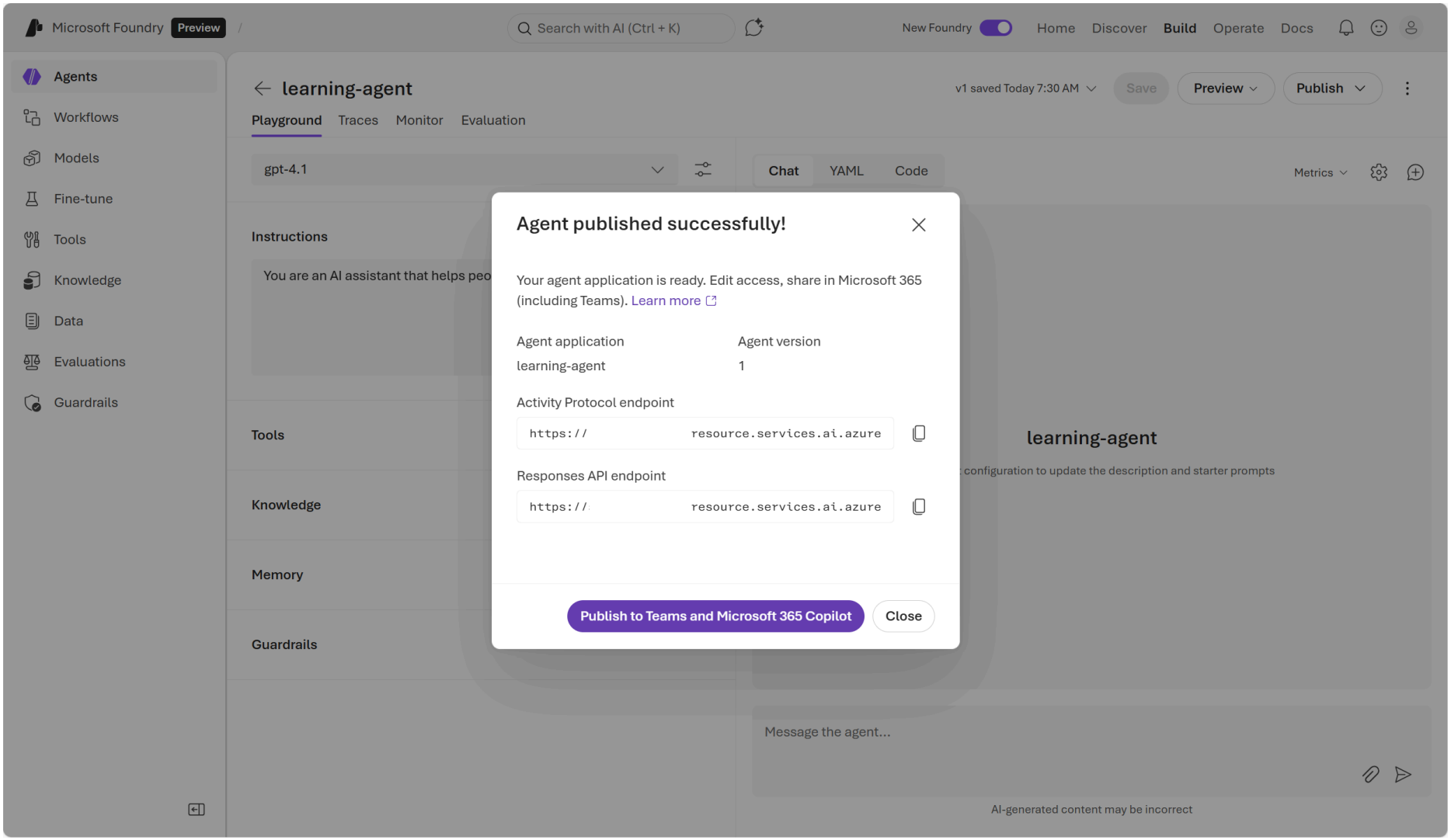
Models
Fine-tune
Tools
Knowledge
Data
Evaluations
Guardrails
learning-agent
v1 saved Today 7:30 AM
Save
Preview
Publish
Playground
Traces
Monitor
Evaluation
gpt-4.1
Chat
YAML
Code
Metrics
Instructions
You are an AI assistant that helps peo
Tools
Knowledge
Memory
Guardrails
Make your agent production-ready
Publish your agent as an application to make it ready for real-world use. You can share it as a Microsoft 365 agent (including Teams). Once published, your agent will have its own URL, identity, and access controls, along with configuration and registration in the Entra Agent Registry. Learn more
Publish target version:
v1
Last saved: Today 7:30 AM
Don't show this message again
Publish
Cancel
learning-agent
configuration to update the description and starter prompts
Message the agent...
AI-generated content may be incorrect



Publishing an agent gives you a stable endpoint to integrate the agent into applications.

Next unit: Exercise - Get started with generative AI and agents in Microsoft Foundry

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